

Problem 2: Properties of the Von Neuman Entropy

h) Solution

Show that:

$$S\left(\sum_i p_i \rho_i\right) \geq \sum_i p_i S(\rho_i)$$

Let's consider a joint state:

$$\rho_{AB} = \sum_i p_i \rho_i^A \otimes |\psi_i\rangle\langle\psi_i|^B$$

$$S_A = -\text{Tr}_B [\rho_{AB} \log \rho_{AB}]$$

$$= -\text{Tr}_B \left[\sum_i p_i \rho_i^A \otimes |\psi_i\rangle\langle\psi_i|^B \right]$$

$$= \sum_i p_i \rho_i^A \text{Tr}_B [|\psi_i\rangle\langle\psi_i|^B]$$

$$= \sum_i p_i \rho_i^A \underbrace{\langle\psi_i|\psi_i\rangle_B}_{=1}$$

$$= \underline{\underline{\sum_i p_i \rho_i^A}}$$

$$S_B = -\text{Tr}_A [\rho_{AB} \log \rho_{AB}]$$

$$= -\text{Tr}_A \left[\sum_i p_i \rho_i^A \otimes |\psi_i\rangle\langle\psi_i|^B \right]$$

$$= \sum_i p_i |\psi_i\rangle\langle\psi_i|^B \underbrace{\text{Tr}_A [\rho_i^A]}_{=1}$$

$$= \sum_i p_i |\psi_i\rangle\langle\psi_i|^B \cdot 1$$

$$= \underline{\underline{\sum_i p_i |\psi_i\rangle\langle\psi_i|^B}}$$

Calculate the Entropy

$$S(B) = S(\rho_B)$$

$$= S\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|^B\right)$$

$$= H(p_1, p_2, \dots)$$

$$= \underline{\underline{H(p)}}$$

$$S(A) = S\left(\sum_i p_i \rho_i^A\right)$$

$$\underline{\underline{\quad\quad\quad}}$$

$$S(A_P) = -\sum_{i,k} p_i \lambda_{i,k} \log \lambda_{i,k}$$

$$= -\sum_{i,k} p_i \lambda_{i,k} \underbrace{\log(\lambda_{i,k})}_{= \log p_i + \log d_{i,k}}$$

$$= -\sum_{i,k} p_i \lambda_{i,k} (\log p_i + \log d_{i,k})$$

$$= -\sum_{i,k} p_i \lambda_{i,k} \log p_i - \sum_{i,k} p_i \lambda_{i,k} \log d_{i,k}$$

There are two terms

$$\sum_k \lambda_{i,k} = 1$$

The first term

$$-\sum_{i,k} p_i \lambda_{i,k} \log p_i = -\sum_{i,k} p_i \log p_i \underbrace{\lambda_{i,k}}_{=1}$$

$$= -\sum_i p_i \log p_i$$

$$= \underline{\underline{H(P)}}$$

The second term

$$-\sum_{i,k} p_i \lambda_{i,k} \log d_{i,k} = \sum_i p_i \left(-\sum_k \lambda_{i,k} \log d_{i,k} \right)$$

$$= \underline{\underline{\sum_i p_i S(P_i)}}$$

$$S(A_P) = H(P) + \sum_i p_i S(P_i) //$$

19917 subadditivity

$$S(AB) \leq S(A) + S(B)$$

$$H(\mathcal{P}) + \sum_i p_i S(\mathcal{B}_i) \leq S\left(\sum_i p_i \mathcal{B}_i\right) + H(\mathcal{P})$$

$$H(\mathcal{P}) - H(\mathcal{P}) + \sum_i p_i S(\mathcal{B}_i) \leq S\left(\sum_i p_i \mathcal{B}_i\right)$$

$$\sum_i p_i S(\mathcal{B}_i) \leq S\left(\sum_i p_i \mathcal{B}_i\right)$$

2) Solution

Show that:

$$S\left(\sum_i p_i \mathcal{B}_i\right) = \sum_i p_i S(\mathcal{B}_i)$$

$$\mathcal{B}AB = \sum_i p_i \mathcal{B}_i \otimes |p_i\rangle\langle p_i|_B$$

$$S(AB) = H(\mathcal{P}) + \sum_i p_i S(\mathcal{B}_i)$$

"only $\mathcal{P}$ ": $\mathcal{P}$ and all $\mathcal{B}_i$ is identical

$$S\left(\sum_i p_i \mathcal{B}_i\right) = \sum_i p_i S(\mathcal{B}_i)$$

$$S(A) = \sum_i p_i S(\mathcal{B}_i)$$

$$\begin{aligned} S(AB) &= \underbrace{H(\mathcal{P})}_{= S(B)} + \underbrace{\sum_i p_i S(\mathcal{B}_i)}_{= S(A)} \\ &= \end{aligned}$$

$$S(AB) = S(A) + S(B)$$

Von Neumann Entropy

$$\mathcal{B}_{AB} = \mathcal{B}_A \otimes \mathcal{B}_B$$

$$= \left(\sum_k p_k \mathcal{B}_k \right) \otimes \left(\sum_j p_j |\psi_j\rangle\langle\psi_j| \right)$$

$$= \sum_{k,j} p_k p_j \mathcal{B}_k \otimes |\psi_j\rangle\langle\psi_j|$$

Two operators are equal, their blocks must be equal to every j

$$\mathcal{B}_k \mathcal{B}_j = p_j \left(\sum_k p_k \mathcal{B}_k \right), \forall j$$

$$\mathcal{B}_j = \sum_k p_k \mathcal{B}_k, \forall j$$

So, all $\mathcal{B}_j$ must be the same state

$$\mathcal{B}_1 = \mathcal{B}_2 = \dots = \mathcal{B}_i = \dots = \bar{\mathcal{B}}$$

If all $\mathcal{B}_i$ identical $\Rightarrow$ equilibrium

$$\mathcal{B}_i = \bar{\mathcal{B}} \quad \forall i$$

$$\begin{aligned} \sum_i p_i \mathcal{B}_i &= \sum_i p_i \bar{\mathcal{B}} \\ &= \bar{\mathcal{B}} \end{aligned}$$

So,

$$S\left(\sum_i p_i \mathcal{B}_i\right) = S(\bar{\mathcal{B}})$$

On other hand

$$\sum_i p_i S(\mathcal{B}_i) = \sum_i p_i S(\bar{\mathcal{B}}) = \underbrace{\left(\sum_i p_i\right)}_{=1} S(\bar{\mathcal{B}}) = S(\bar{\mathcal{B}}) \quad \text{Thus}$$

$$\text{give,} \quad S\left(\sum_i p_i \mathcal{B}_i\right) = \sum_i p_i S(\mathcal{B}_i)$$

3) Solution

$$S(X) = S(Y)$$

- for pure bipartite state $\rightarrow$ equal entropies.

$$|\psi\rangle_{XY} = \sum_k \sqrt{\lambda_k} |k\rangle_X \otimes |k\rangle_Y$$

with $\lambda_k \geq 0$

$$\sum_k \lambda_k = 1$$

$$\rho_X = \text{tr}_Y [|\psi\rangle\langle\psi|]$$

$$= \sum_k \lambda_k |k\rangle\langle k|_X$$

$$\rho_Y = \text{tr}_X [|\psi\rangle\langle\psi|]$$

$$= \sum_k \lambda_k |k\rangle\langle k|_Y$$

$$S(X) = S(\rho_X)$$

$$= - \sum_k \lambda_k \log \lambda_k$$

$$= - \sum_k \lambda_k \log \lambda_k$$

$$S(Y) = S(\rho_Y)$$

$$= - \sum_k \lambda_k \log \lambda_k$$

$$= - \sum_k \lambda_k \log \lambda_k$$

$$\underline{\underline{S(X) = S(Y)}}$$

Apply to $|\psi\rangle_{ABCD}$

$|\psi\rangle_{ABCD} \in \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C \otimes \mathcal{H}_D$ is pure

a) Bisection: A vs BCD

$$\underline{H_A} \otimes (H_B \otimes H_C \otimes H_D), \quad A|BCD \rightarrow$$

$$S(A) = S(BCD) \rightarrow \text{like the above}$$

2) Bipartition AB vs CD

$$(H_A \otimes H_B) \otimes (H_C \otimes H_D), \quad AB|CD$$

$$S(AB) = S(CD)$$

3) Bipartition ABC vs D

$$(H_A \otimes H_B \otimes H_C) \otimes H_D, \quad ABC|D$$

$$S(ABC) = S(D)$$

Therefore,

pure state of any number of particles, for every bipartition, the von Neumann Entropies of the two blocks are equal

$$\left. \begin{aligned} S(A) &= S(BCD), \\ S(AB) &= S(CD), \\ S(ABC) &= S(D) \end{aligned} \right\} \text{always true,}$$

4) Joining

$$S(A|B) = S(AB) - S(B)$$

$$S(A|BC) = S(ABC) - S(BC)$$

$$S_{ABC} = S_{AB} \otimes S_C$$

Entropy of the product state

$$\begin{aligned} S(ABC) &= S(S_{AB} \otimes S_C) \\ &= \underline{S(AB) + S(C)} \end{aligned}$$

The reduced state

$$\begin{aligned} S_{BC} &= \text{Tr}_A[S_{AB} \otimes S_C] \\ &= \underline{S_B \otimes S_C} \end{aligned}$$

$$\begin{aligned} S(BC) &= S(S_{BC}) \\ &= S(S_B \otimes S_C) \\ &= \underline{S(B) + S(C)} \end{aligned}$$

Compute, $S(A|BC)$

$$\begin{aligned} S(A|BC) &= \underbrace{S(ABC)}_{= S(AB) + S(C)} - \underbrace{S(BC)}_{= S(B) + S(C)} \end{aligned}$$

$$\begin{aligned} S(A|BC) &= S(AB) + \cancel{S(C)} - S(B) - \cancel{S(C)} \\ &= S(AB) - S(B) \\ &= \underline{S(A|B)} \end{aligned}$$

$$S(A|BC) = S(A|B)$$

Problem 2: Quantum relative information

1) solution

$$P_A = \sum_i p_i |i\rangle\langle i|_A, \quad p_i \geq 0$$

$$Q_B = \sum_j q_j |j\rangle\langle j|_B, \quad q_j \geq 0$$

$$P_A \otimes Q_B = \sum_{i,j} p_i q_j |i\rangle\langle i|_A \otimes |j\rangle\langle j|_B$$

operation $\log(P_A \otimes Q_B)$:

$$\log(P_A \otimes Q_B) = \sum_{i,j} \log(p_i q_j) |i\rangle\langle i|_A \otimes |j\rangle\langle j|_B.$$

$$= \sum_{i,j} (\log p_i + \log q_j)$$

$$= \sum_{i,j} (\log p_i + \log q_j) |i\rangle\langle i|_A \otimes |j\rangle\langle j|_B$$

$$= \sum_{i,j} \log p_i |i\rangle\langle i|_A \otimes |j\rangle\langle j|_B + \sum_{i,j} \log q_j |i\rangle\langle i|_A \otimes |j\rangle\langle j|_B$$

$$= \sum_i \log p_i |i\rangle\langle i|_A \otimes \left(\sum_j |j\rangle\langle j|_B \right) + \left(\sum_i |i\rangle\langle i|_A \right) \otimes \left(\sum_j \log q_j |j\rangle\langle j|_B \right)$$

$\underbrace{\sum_j |j\rangle\langle j|_B}_{=I_B} \quad \underbrace{\sum_i |i\rangle\langle i|_A}_{=I_A}$

$$= \log(P_A) \otimes I_B + I_A \otimes \log(Q_B)$$
